

Week 1 Logic Test

The Case of the Three Bells

A Whately Puzzle on What Follows From What

Name: _____

Date: _____

Whately's test: Find the conclusion. Find the stated reason. Bring the hidden assumption into the light. Then ask: does the conclusion really follow? Remember also what logic cannot do: logic can test reasoning, but it cannot supply missing facts, wisdom, virtue, observation, or testimony.

The Puzzle Story

At the Academy of Straight Thinking, three bells hang in the old tower.

- The **Silver Bell** is supposed to be rung when a student has spoken wisely.
- The **Brass Bell** is rung when a student has spoken confidently.
- The **Iron Bell** is rung when a student has won loud applause.

One morning, the class finds this note pinned to the tower door:

“Whoever rang the bell last night must be the wisest student in the Academy.”

Three students are questioned.

Clara says: “I rang the Iron Bell. Everyone applauded my speech yesterday, so my argument must have been true.”

Miles says: “I rang the Brass Bell. I spoke more confidently than anyone else, so I must have had the strongest proof.”

Jonas says: “I rang the Silver Bell. The rules say the Silver Bell is for wisdom, so if I rang it, I must have spoken wisely.”

Then the Head Tutor adds:

“Be careful. Bells may tell us something, but they may not tell us everything. Logic can test whether a conclusion follows, but it cannot discover facts we have not yet learned.”

Part I: Diagnose the Three Claims

For each student, identify the parts of the argument and test whether the conclusion follows.

1. Clara and the Iron Bell

Claim: “Everyone applauded my speech yesterday, so my argument must have been true.” **Conclusion:**

Stated reason:

Hidden assumption:

Does the conclusion follow? Explain briefly.

2. Miles and the Brass Bell

Claim: “I spoke more confidently than anyone else, so I must have had the strongest proof.” **Conclusion:**

Stated reason:

Hidden assumption:

Does the conclusion follow? Explain briefly.

3. Jonas and the Silver Bell

Claim: “The rules say the Silver Bell is for wisdom, so if I rang it, I must have spoken wisely.” **Conclusion:**

Stated reason:

Hidden assumption:

Does the conclusion follow? Explain briefly.

Part II: Rank the Reasoning

Answer in complete thoughts.

1. Which student’s reasoning is weakest? Why?

2. Which student’s reasoning is strongest or least weak? Why is it still not complete proof?

3. What fact or testimony would the class still need before deciding who actually deserved the Silver Bell?

Part III: What Can Logic Decide?

Mark each item as **Logic can test this** or **Logic alone cannot decide this**. Then give a short reason for two of your choices.

1. Whether applause proves that an argument is true.
2. Whether confidence proves that a speaker has strong proof.
3. Whether Jonas actually rang the Silver Bell honestly and correctly.
4. Whether a conclusion follows from the reason given.
5. Whether a student is wise or virtuous in character.

Explain two choices:

Part IV: Repair One Argument

Choose Clara, Miles, or Jonas. Rewrite the student’s argument so that it becomes more careful. You may narrow the conclusion, add needed evidence, or admit what has not yet been proved.

Student chosen:

Repaired argument:

Part V: Macbeth Bridge

Macbeth hears a prophecy. One part of it comes true. Then he begins to imagine kingship and murder.

Complete the comparison:

Macbeth's dangerous reasoning is like the bell puzzle because he is tempted to treat _____ as if it proves _____.

Now explain your answer in 4–6 sentences. Use the words *conclusion*, *reason*, *hidden assumption*, and *follows*.

Final Whately Claim

In one sentence, state what this puzzle teaches about what logic can and cannot do.
